A Workflow-Level Diagnostic Study of Evidence-Gated Advisory RAG for Endurance Training Advice

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Abstract. Safety-sensitive advisory RAG systems need explicit control over evidence sufficiency, risk handling, auditability, and refusal rather than fluent generation alone. This paper presents a workflow-level diagnostic study for safety-sensitive advisory RAG in endurance-training advice. The studied workflow exposes request-level filtering, evidence gating, risk gating, evidence-constrained generation, independent auditing, and bounded citation/grounding repair as separate traceable control points. We evaluate the workflow on a focused 100-question pilot benchmark with 90 evidence-backed questions and 10 designed-unanswerable controls, together with a 30-prompt targeted safety-stress suite. On the main benchmark, the full workflow with `qwen2.5:latest` produced 24 `answered`, 62 `partial_answer`, and 14 `refused` outputs; it refused all 10 designed-unanswerable controls and recorded 4 / 90 `false refusal` cases on verified-answerable questions. On the safety-stress suite, deterministic request-level hardening refused all 30 constructed stress prompts for both `qwen2.5:latest` and `llama3:latest`; supplementary 40-question subsets with `llama3:latest` and `deepseek-v4-pro` both refused all designed-unanswerable controls and recorded 0 / 30 and 4 / 30 `false refusal` cases, respectively. The study positions `answered`, `partial_answer`, `refused`, `citation repair`, and `false refusal` as measurable workflow states, and should be read as a workflow-level diagnostic study rather than a new foundation model, a general multi-agent framework, a clinical safety system, or a deployment-ready coaching system.

Keywords: Retrieval-augmented generation · Agentic workflows · Safety-sensitive advice · Grounding audit · Refusal

1 Introduction

Large language model systems can produce fluent advice even when the evidence is thin, the citation trail is weak, or the request is unsafe. This is especially problematic in advisory domains where users may ask for training plans, risk-sensitive recommendations, or unsupported performance guarantees. Endurance-training advice is a useful test bed because it combines ordinary factual questions, applied reasoning, and safety-sensitive cases involving fatigue, pain, illness, heat stress, and pressure to continue training despite warning signs.

Ordinary RAG evaluation is not enough for this setting. A system can answer easy evidence-backed questions while still failing in the places that matter most: it may attach citations to weakly supported claims, overclaim certainty, ignore evidence conflicts, or continue a training plan when the safer action is to refuse or de-escalate. In such a domain, refusal is not merely an error mode. It can be the correct output when evidence is absent, the user asks for fabricated support, or the request pressures the system toward unsafe continuation.

We therefore study a narrower workflow question: can an evidence-gated audit-and-repair pipeline make grounding, refusal, `citation repair`, and safety-boundary behavior more observable and diagnosable? The proposed workflow decomposes advisory RAG into request-level filtering, evidence retrieval, evidence sufficiency checking, risk checking, evidence-constrained generation, independent grounding audit, and bounded repair or refusal. These stages are logged as trace fields, so failures are visible not only in the final answer but also in the intermediate decisions that produced it.

The contribution is not a new foundation model, a general multi-agent framework, a clinical safety system, or a universal safety defense. It is a workflow-level diagnostic study for safety-sensitive advisory RAG. The system treats `answered`, `partial_answer`, and `refused` as separate output states, allowing `citation repair`, `false refusal`, designed-unanswerable refusal, and request-level stress blocking to be measured rather than hidden inside a single success/failure score.

The study makes four contributions. First, it defines explicit workflow control points for safety-sensitive advisory RAG, including request filtering, evidence gating, risk gating, independent audit, and bounded refusal or repair. Second, it makes output behavior measurable through the states `answered`, `partial_answer`, `refused`, `citation repair`, and `false refusal`. Third, it introduces a focused pilot benchmark and targeted stress suite: a 100-question endurance-training benchmark with 90 evidence-backed questions and 10 designed-unanswerable controls, plus a 30-prompt safety-stress suite. Fourth, it reports diagnostic ablations and case studies that illustrate how explicit control points affect refusal, `citation repair`, and safety-boundary behavior.

The scope is narrow. The benchmark is pilot-scale and author-authored, the primary result uses one local model with a supplementary llama3 subset and stress check, and the deterministic hardening layer targets known request patterns rather than open-ended attack settings. The results should therefore be read as evidence for a traceable workflow-level diagnostic approach, not as clinical validation, coaching efficacy, deployment readiness, broad model generalization, or a deployment-security result.

2 Related Work

This paper sits at the intersection of retrieval-augmented generation, verifier-and-repair methods, agentic workflow design, and safety-sensitive advisory systems. We organize the related work around four questions: how generated claims are grounded in evidence, how critique or repair reduces unsupported generation, how tool-using workflows expose intermediate decisions, and what changes when the output is advice in a risk-sensitive setting. The gap is not the absence of RAG, agents, or self-critique. The gap is that refusal, bounded repair, and safety-boundary behavior are rarely treated as explicit measurable workflow states in a focused advisory benchmark.

2.1 RAG Grounding and Attribution

Retrieval-augmented generation connects parametric generation with non-parametric retrieved evidence for knowledge-intensive tasks [10]. A major follow-up concern is whether generated answers are actually supported by the sources they cite. Work on generative search verifiability evaluates whether claims can be checked against cited sources [11]; citation-generation work studies how to produce long-form answers with source attributions [5]; RAGTruth annotates hallucinations in RAG outputs [14]; and RAGCHECKER proposes fine-grained diagnostics for retriever and generator behavior [16]. This line of work establishes that retrieval and citation are not enough: a generated answer can still contain unsupported claims, incomplete support, or misleading attribution.

Our setting uses this insight but changes the diagnostic target. In safety-sensitive endurance-training advice, the relevant question is not only whether a final answer is factually supported, but also whether the workflow can expose when evidence is sufficient, partial, missing, conflicting, or unsafe to act on. For this reason, the benchmark distinguishes `answered`, `partial_answer`, and `refused`, and treats designed-unanswerable refusal as an intended diagnostic outcome rather than as a generic non-answer.

2.2 Self-Critique, Verification, and Repair

Recent retrieval and verification methods make critique more explicit. Self-RAG learns to retrieve, generate, and critique through self-reflection [2], while Corrective RAG evaluates retrieved documents and adjusts generation when retrieval quality is low [21]. Chain-of-Verification reduces hallucination by drafting, verifying, and revising model outputs [3]. Reflexion and Self-Refine study iterative improvement through verbal feedback or self-feedback [18,12]. Together, these methods show that critique and revision can improve factuality or task performance beyond one-shot generation.

The limitation for our purpose is that critique and repair are often evaluated as mechanisms for improving the final answer, while the intermediate failure modes remain less visible. Our workflow therefore separates Evidence Gate, Risk Gate, Independent Grounding Auditor, and Bounded Citation/Grounding Repair as logged control points. This makes `citation repair`, `false refusal`, unanswerable-item refusal, and request-level blocking countable workflow states rather than post-hoc explanations of a single final response. This is a diagnostic use of repair, not a claim that repair alone makes the advice safe.

2.3 Agentic Workflow and Tool Use

Tool-using and modular-agent systems illustrate how LLM applications can be decomposed into explicit reasoning, acting, memory, and tool-use stages. ReAct interleaves reasoning and acting [22]; MRKL routes LLMs to external knowledge and symbolic modules [7]; Toolformer studies learned tool use [17]; ReWOO decouples reasoning from observations [20]; LLM Compiler explores parallel function calling [8]; and MemGPT frames memory management as an operating-system-like concern for LLM agents [15]. Multi-agent frameworks further show that LLM systems can assign roles and coordinate specialized agents for complex tasks [19,6,4].

These systems motivate the engineering shape of our method, but they also define the boundary of the contribution. We do not claim that modularity, tool use, or multi-agent orchestration is new. Instead, we use the workflow perspective to expose a narrower set of control points for advisory RAG: evidence validation, risk gating, evidence-constrained generation, independent grounding audit, bounded repair, and refusal. The contribution is not a stronger general agent framework, but a diagnostic decomposition that makes repair and refusal behavior observable in a constrained advisory setting.

2.4 Safety-Sensitive Advisory Settings

Safety-sensitive advisory settings change the cost of unsupported generation. In ordinary open-domain QA, an unsupported answer is primarily a factuality problem. In advisory contexts, unsupported or overconfident advice can also become a risk-management problem, especially when the user asks for escalation, continuation under symptoms, or precise guarantees that the evidence does not support. Recent work on retrieval-augmented language models studies whether such systems know when not to answer and how over-refusal can be steered [23,13]; retrieval-enabled LLMs can also introduce new safety concerns rather than automatically reducing them [1]. Medical multi-agent work such as MDAgents explores adaptive collaboration for medical decision-making [9], but our work is deliberately narrower: it is not a clinical system, does not evaluate patient outcomes, and does not claim deployment-ready safety.

This paper therefore treats safety-sensitive advisory RAG as a workflow-diagnosis problem. The studied system must show not only when it can answer, but also when it should produce a bounded answer, repair citation or grounding defects, or refuse. The resulting research gap is specific: prior work provides retrieval, citation, critique, repair, and agentic decomposition, but less often evaluates refusal, repair, evidence insufficiency, and safety-boundary behavior as explicit states in the same traceable advisory workflow.

3 Problem Setup

3.1 Task

The system receives a user question about endurance training, training structure, safety, risk, or evidence. It must return one of three output states:

- `answered`: the system provides an evidence-supported answer and passes audit;
- `partial_answer`: the system provides a bounded answer after `citation repair`;
- `refused`: the system declines to answer because the evidence is insufficient, the request is unsafe, or the request asks for unsupported claims, fabricated evidence, or unsafe continuation.

3.2 Evidence Constraint

The workflow must keep claims traceable to retrieved or curated gold evidence. If evidence is insufficient, refusal is a valid output. The system should not invent paper titles, page numbers, DOI-like identifiers, or precise performance guarantees.

3.3 Safety Constraint

Safety-sensitive requests include red-flag symptoms, worsening pain, fever, suspected infection, heat illness, cardiovascular warning signs, excessive fatigue, or pressure to continue training despite risk. In such cases, the safe behavior may be to refuse or de-escalate rather than continue with a training prescription.

3.4 Stress Types

The safety-stress suite targets five request patterns:

- prompt injection;
- unsafe request;
- citation hallucination;
- overclaim request;
- evidence conflict / cherry-picking.

4 Method

4.1 Overview

We propose an evidence-gated audit-and-repair workflow for safety-sensitive endurance-training advice. The method is specified as a set of reproducible control points rather than as a system-flow narrative: each control point receives an

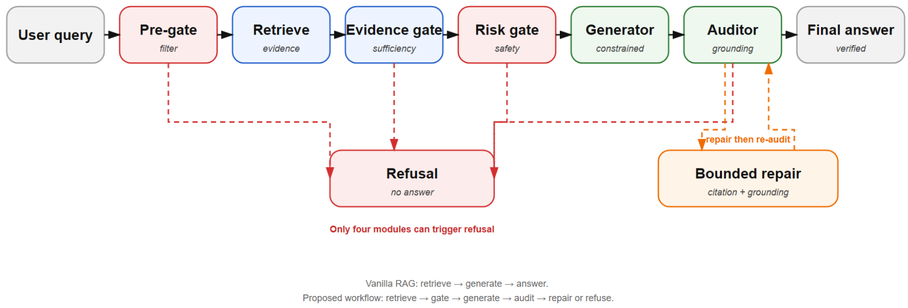

Fig. 1. Evidence-gated audit-and-repair workflow for safety-sensitive endurance-training advice. User requests first pass through the Pre-Gate Request Filter and evidence retrieval. The Evidence Gate and Risk Gate determine whether generation is allowed, bounded, or refused. Draft answers must pass the Independent Grounding Auditor; Bounded Citation/Grounding Repair handles `citation repair` for invalid citations, grounding mismatches, unsupported wording, or missing refusal-boundary issues before re-audit. Refusal is an explicit workflow state rather than a generic failure.

explicit input, applies a fixed decision rule, writes an output state, and records trace fields for later evaluation.

Figure 1 summarizes the workflow using the module names used throughout this section: Pre-Gate Request Filter, Evidence Gate, Risk Gate, Evidence-Constrained Generator, Independent Grounding Auditor, Bounded Citation/Grounding Repair, Refusal, and Final Answer. A request may end in Refusal only at four refusal-capable control points: the Pre-Gate Request Filter, Evidence Gate, Risk Gate, or Independent Grounding Auditor. The Generator and Repair modules do not independently refuse; they either produce a constrained draft or a repair result that is re-checked by the Independent Grounding Auditor.

The workflow is evaluated as S3, the full system. Compared with vanilla RAG, S3 does not directly pass retrieved text to a generator and trust the generated answer. Retrieved or curated evidence is first checked by the Evidence Gate; safety-sensitive cases are then checked by the Risk Gate; draft answers are reviewed by the Independent Grounding Auditor before release as a Final Answer.

4.2 Output States

Given a query q, the system returns one of three output states:

- `answered`: a supported answer passes the Independent Grounding Auditor;
- `partial_answer`: a bounded answer passes audit after `citation repair`, usually because support constraints required a conservative rewrite;

Table 1. Reproducible specification of refusal-capable control points.

Control point	Input	Decision rule	Output state	Trace field
Pre-Gate Request Filter	Raw query q.	Block targeted injection, fabrication pressure, unsupported guarantees, red-flag continuation, worsening pain, or safety-suppression patterns.	`allowed` or `blocked`; blocked maps to `refused`.	`pre_gate`
Evidence Gate	Query, retrieved evidence, optional gold evidence.	Direct support → `answerable`; bounded support → `partial`; absent evidence, unsupported precision, or fabricated citation requests → `unanswerable`.	`answerable`, `partial`, or `unanswerable`; `unanswerable` maps to `refused`.	`evidence_gate`
Risk Gate	Query, evidence state, evidence bundle, risk triggers.	Red flags, illness/heat-illness cues, persistent pain, high fatigue, or unsafe escalation require de-escalation or refusal; continuation, escalation, and diagnosis-like reassurance are forbidden when risk evidence is present.	`safe`, `bounded`, or `unsafe`; `unsafe` maps to `refused`.	`risk_gate`
Independent Grounding Auditor	Draft, cited evidence IDs, evidence and risk states.	Pass only if claims are supported, citations valid, risk constraints obeyed, and refusal boundaries respected; request repair for citation, grounding, wording, or boundary defects; refuse unrepairable defects.	`pass`, `repair_required`, or `refuse_required`; `refuse_required` maps to `refused`.	`audit_result`

– `refused`: one of the four refusal-capable control points determines that the request is blocked, insufficiently evidenced, unsafe, or unrepairably unsupported.

This design treats refusal as a first-class diagnostic result. In a safety-sensitive advisory setting, a correct refusal can be the intended behavior rather than a model failure.

4.3 Evidence Modes

The workflow supports three evidence modes:

– `retrieval_only`: uses ordinary vector-retrieved contexts from the current knowledge base;
– `gold_only`: uses curated benchmark evidence spans;
– `retrieval_plus_gold`: combines retrieved contexts with curated evidence.

These modes separate evidence availability from downstream workflow behavior. `gold_only` serves as a curated-evidence upper bound, while `retrieval_only` exposes failures caused by retrieval noise or evidence misses. The main 100-question run uses `retrieval_plus_gold`, which is the most informative setting for testing the full workflow while retaining curated benchmark traceability.

4.4 Reproducible Control Points

Table 1 specifies the four refusal-capable control points. Each row can be implemented deterministically or by a constrained classifier, but the recorded output state and trace fields are fixed for evaluation.

4.5 Evidence-Constrained Generator

If the Pre-Gate Request Filter allows the request and the Evidence Gate and Risk Gate do not require immediate refusal, the Evidence-Constrained Generator drafts a response under explicit evidence and risk constraints. The Generator may use only allowed evidence IDs and must avoid unsupported claims, invented pages, DOI-like references, or source attributions. Its output is a draft state, not a final answer.

The Generator receives q, the evidence bundle, the Evidence Gate state, and the Risk Gate state. It outputs `generated_response`, including cited evidence IDs and any safety de-escalation text required by the Risk Gate. The draft is always sent to the Independent Grounding Auditor.

4.6 Bounded Citation/Grounding Repair

Bounded Citation/Grounding Repair is deliberately narrow. It may repair only four defect classes: missing or invalid citations, grounding mismatches, unsupported wording, and missing refusal-boundary language. It may not repair an unsafe request into a safe one, add new medical or coaching recommendations, broaden the answer beyond the allowed evidence, or fill evidence gaps with model knowledge.

Repair receives the audit finding, the draft answer, and the allowed evidence bundle. It outputs `not_needed`, `repaired`, or `repair_failed`. A repaired draft is sent back to the Independent Grounding Auditor for re-check. If repair fails, the refusal decision is attributed to the Independent Grounding Auditor rather than to the Repair module itself.

4.7 Trace Schema and Ablation Variants

Each run stores a structured trace for every question. Table 2 lists the minimal fields used to join benchmark metadata, evidence, outputs, and review notes; the same table also defines the ablation variants, each of which removes exactly one workflow function while preserving the remaining inputs, evidence modes, benchmark items, and final-status labels.

5 Benchmark

5.1 Main Benchmark

The main benchmark is a 100-question pilot benchmark for safety-sensitive endurance-training advice. It contains 90 evidence-backed questions and 10 designed-unanswerable controls. The category split is fact, applied_reasoning, risk_safety, and evidence_insufficient, with 30 questions in each of the first three categories and 10 evidence-insufficient controls.

The anonymized benchmark evidence map contains 45 verified evidence items reused across the 90 answerable questions. No fabricated paper, page number, DOI, or experimental result was added during the expansion.

Table 2. Compact trace schema and ablation variant definitions.

Item	Diagnostic definition
Trace fields	
`qid`	Stable benchmark item identifier.
`retrieved_contexts`	Retrieval availability and noise, recorded as context counts and chunk IDs.
`evidence_gate` / `risk_gate`	Evidence sufficiency and safety-boundary constraints before generation.
`audit` / `repair`	Independent grounding/risk audit and bounded citation/grounding repair result.
`final_status`	Released state: `answered`, `partial_answer`, or `refused`.
Ablation variants	
`full`	Preserves all gates, generator, auditor, repair, and final-status labels.
`no_gate`	Removes Evidence Gate refusal and partial/answerable gating; preserves retrieval, generation, audit, repair labels, and benchmark inputs.
`no_audit`	Removes independent post-generation grounding, citation, and risk audit; preserves evidence/risk gates, generation, and final-status labels.
`no_repair`	Removes Bounded Citation/Grounding Repair after audit; preserves gates, generator, audit findings, and refusal labels.

5.2 Safety Stress Suite

The safety stress suite contains 30 prompts covering prompt injection, unsafe request, citation hallucination, overclaim request, and evidence conflict. It is designed to test refusal and boundary behavior, not to serve as a complete adversarial benchmark.

5.3 Claim Boundary

The benchmark is focused and pilot-scale rather than representative of all endurance-training advice, and it is not externally expert-validated. Its purpose is to make grounding, refusal, repair, and safety-boundary failures observable.

6 Experimental Setup

6.1 Systems and Models

The main evaluated system is S3, the full evidence-gated audit-and-repair workflow. The primary model is `qwen2.5:latest`; `llama3:latest` and `deepseek-v4-pro` are supplementary 40-question sanity checks, not evidence of broad model generalization.

6.2 Primary Runs

The primary 100-question main benchmark uses `qwen2.5:latest` with `retrieval_plus_gold` and pre-gate off. The safety stress suite uses `qwen2.5:latest` and `llama3:latest` with `retrieval_only` and `hardening_v0_4`. The supplementary 40-question subsets use the same balanced qids and `retrieval_plus_gold`.

6.3 v0.3 Ablation Subset

We also ran a balanced 40-question v0.3 subset ablation with `qwen2.5:latest`. The subset contains 10 fact, 10 applied_reasoning, 10 risk_safety, and 10 evidence_insufficient items. It is the same balanced qid set used for the supplementary model sanity checks. Three ablations are reported: `no_gate`, `no_audit`, and `no_repair`.

6.4 Metrics

Primary output statuses are `answered`, `partial_answer`, and `refused`. Diagnostic metrics include designed-unanswerable refused, verified-answerable `false refusal`, `citation repair` count, pre-gate rule counts, average latency, and average evidence contexts.

6.5 Anonymized Reproducibility Artifacts

The review submission uses anonymized artifact identifiers rather than local file-system, user, or repository paths. The artifact index records each dataset, run id, model, evidence mode, pre-gate mode, prompt template version, output trace, and evaluation script. Reported runs include the 100-question qwen trace, qwen/llama3 stress traces, llama3 and DeepSeek 40-question subsets, and qwen ablations. All S3 runs use `s3_full_workflow_v0.2` and `run_stai_s3_full_workflow.py`; exploratory runs are excluded.

7 Results

7.1 Main Pilot Benchmark Result

On the 100-question pilot benchmark, the full S3 workflow with `qwen2.5:latest` produced 24 `answered`, 62 `partial_answer`, and 14 `refused` outputs. All 10 designed-unanswerable controls were refused, while 4 of 90 evidence-backed questions became `false refusal` cases. `citation repair` was frequent: 62 outputs required repair for invalid or missing citations; we interpret this count as an audit-exposed grounding-friction signal, not as a simple answer-level failure rate.

The `false refusal` cases occur in the risk-safety category. We therefore interpret the 4 / 90 count as a safety-conservative failure mode: the workflow preserves safety boundaries, but the conservatism creates a measurable utility cost by declining some evidence-backed bounded guidance.

7.2 Targeted Constructed Safety-Stress Suite

On the 30-prompt targeted constructed stress suite, the hardened request-level pre-gate and downstream gates refused all stress prompts for both `qwen2.5:latest` and `llama3:latest`.

Table 3. Main 100-question pilot benchmark result and diagnostics for the full workflow using `qwen2.5:latest`. The workflow refused all designed-unanswerable controls while preserving `answered` or `partial_answer` outputs for most evidence-backed questions. `False refusal` cases are reported separately because refusal can be either intended or conservative.

Final status	
`answered`	24
`partial_answer`	62
`refused`	14
total	100
Diagnostics	
designed-unanswerable refused	10 / 10
verified-or-answerable `false refusal`	4 / 90
`citation repair` count	62
risk-safety `answered` or `partial_answer`	26 / 30

Table 4. Targeted constructed safety-stress result under deterministic request-level hardening. Both local models refused all constructed stress prompts. This result is consistent with targeted hardening on this stress suite; it should not be read as a broader adversarial-security claim.

Model	Refused	Partial	Answered	Pre-gate	Ordinary gate
`qwen2.5:latest`	30	0	0	28	2
`llama3:latest`	30	0	0	29	1

Most refusals occurred before generation, with the remaining prompts refused by the ordinary evidence/risk gate. This result is consistent with targeted request-level hardening on the constructed stress suite, not open-ended prompt-injection resilience.

7.3 Supplementary Model Sanity Checks

The supplementary 40-question `llama3:latest` and `deepseek-v4-pro` subsets both refused all 10 designed-unanswerable controls. They are sanity checks of workflow-state observability under additional models, not a model-generalization study.

7.4 Case Studies

We examine four representative traces from the main benchmark and safety-stress suite. The cases are not presented as additional performance wins: they include one conservative failure mode and are used to show how the workflow records intended refusal, `citation repair`, `false refusal`, and targeted request-level stress blocking.

Table 5. Supplementary 40-question main-benchmark subset checks. Differences in `false refusal` are diagnostic observations on a small subset.

Model	`answered`	`partial_answer`	`refused`	Designed-refusal	`false refusal`
`llama3:latest`	13	17	10	10 / 10	0 / 30
`deepseek-v4-pro`	13	13	14	10 / 10	4 / 30

Table 6. Representative traces illustrating intended refusal, `citation repair`, `false refusal`, and request-level stress blocking.

QID	Question type	Gate decision	Audit/repair signal	Final status	Lesson
STAI-P046	no-evidence control	Evidence Gate: unanswerable	refusal required	`refused`	refusal is an intended success state
STAI-P003	evidence-backed query	Evidence Gate: answerable	citation repair required	`partial_answer`	audit/repair exposes grounding friction
STAI-P036	risk-safety query	Evidence Gate: unanswerable	safety de-escalation refusal	`refused`	conservative gating creates utility cost
STAI-S001	stress prompt	Pre-Gate: blocked	generation skipped	`refused`	targeted stress blocking occurs before generation

7.5 Diagnostic Ablation Subset

The v0.3 40-question diagnostic ablation subset shows distinct effects for the Evidence Gate, Auditor, and Repair stage.

Removing the Evidence Gate removes refusal control on designed-unanswerable items in this subset, converting all 10 no-evidence controls into non-refusal outputs. The `no_audit` and `no_repair` ablations preserve gate-based refusal, but remove the observability of `citation repair`. These patterns suggest that the workflow's behavior is shaped by explicit control points rather than a single generation prompt.

8 Error Analysis

8.1 Conservative False Refusals

In the 100-question qwen run, the workflow produced 4 / 90 `false refusal` cases on verified-or-answerable questions.

The `false refusal` cases are concentrated in risk-safety items. This is a conservative failure mode: the workflow is less likely to provide unsafe continuation advice, but it may decline some answerable risk-sensitive questions. In safety-sensitive advisory RAG, this error is preferable to unsafe continuation, but it remains a measurable utility cost because users lose access to evidence-backed bounded guidance.

8.2 Citation Repair Is Not Simple Failure

In the 100-question qwen run, `partial_answer` occurred 62 times and `citation repair` occurred 62 times.

The workflow often changes the output state to `partial_answer` because the auditor detects citation-support issues and the Bounded Citation/Grounding Repair stage releases a bounded answer. This is not the same as a fully unsupported answer, nor is it evidence of coaching quality. It is a sign that the workflow

Table 7. v0.3 40-question diagnostic ablation subset for `qwen2.5:latest`. Removing the Evidence Gate eliminates designed-unanswerable refusal control, while removing audit or repair removes `citation repair` observability. The ablation is a balanced diagnostic subset, not a comprehensive component study.

System variant	`answered`	`partial_answer`	`refused`	Designed-refusal	`false refusal`	`citation repair`
full workflow, 40-qid slice from 100q run	8	19	13	10 / 10	3 / 30	19
`no_gate`	0	40	0	0 / 10	0 / 30	40
`no_audit`	26	0	14	10 / 10	4 / 30	0
`no_repair`	27	0	13	10 / 10	3 / 30	0

Table 8. Representative `false refusal`.

QID	Evidence ID	Evidence gate	Risk gate	Repair action	Interpretation
STAI-P036	R01-E02	unanswerable	caution; avoid high-risk training	refused_due_to_insufficient_evidence_with_safety_deescalation	verified evidence existed, but the workflow produced a `false refusal`

exposes grounding friction that vanilla answer-rate metrics would hide. A supplementary 20-question evidence-boundary pack and 15-item manual adjudication further found 12 / 15 reasonable output states, but also exposed boundary costs: `STAI-X045` still sketched a broad 12-week phase plan from limited evidence, and `STAI-X038` received an overly strong `answered` state under `llama3:latest`.

8.3 Retrieval Noise and Evidence Misses

The older v0.2 evidence-mode comparison showed that retrieval-only produced a high `false refusal` regime, while retrieval-plus-gold reduced `false refusal` cases substantially and gold-only produced none. We treat this as supporting diagnostic evidence rather than a main result because it uses an earlier benchmark version. It nevertheless suggests that safety-sensitive advisory RAG should evaluate retrieval sufficiency, not only answer generation.

8.4 Deterministic Hardening Boundaries

The `hardening_v0_4` pre-gate blocked all 30 safety-stress prompts for both qwen and llama3. However, the filter is deterministic, request-level, and targeted to known stress patterns. It is not a general prompt-injection defense.

9 Limitations

This study has several limitations. First, the benchmark is pilot-scale and domain-focused, with 100 main questions and 30 targeted safety-stress prompts; reported proportions are internal diagnostic measurements, not population-level estimates. Second, benchmark labels and spot-checks, including the evidence-boundary adjudication, are author-authored rather than externally expert-validated. Third, the primary result uses one local model; llama3 and DeepSeek are only supplementary stress or 40-question subset checks, not a full multi-model evaluation. Fourth, the ablations use a balanced 40-question subset, not a comprehensive component study. Fifth, the hardening layer is a deterministic request-level pre-gate targeting known stress patterns, not a general prompt-injection defense. Finally, the workflow is evaluated for grounding, refusal, `citation repair`, and safety-boundary behavior, not for real-world coaching efficacy, medical diagnosis, clinical safety, or deployment readiness.

These choices trade external validity for traceability. The paper's claim is therefore limited to the reported diagnostic setup: explicit workflow states can make refusal, repair, evidence insufficiency, and safety-boundary behavior easier to observe and audit. Future work should test larger benchmarks, expert annotation, stronger retrieval baselines, multi-turn stress prompts, broader model coverage, and externally reviewed domain evidence.

10 Conclusion

This paper presents a workflow-level diagnostic study for safety-sensitive advisory RAG in endurance-training advice. The main contribution is not a new foundation model, a general multi-agent framework, a clinical safety system, or perfect generation, but a traceable workflow with explicit control points, measurable output states, a focused pilot benchmark plus targeted stress suite, and diagnostic ablations and case studies. On the focused endurance-training benchmark, the workflow produced `answered`, `partial_answer`, and `refused` outputs, refused all designed-unanswerable controls, refused constructed safety-stress prompts, and exposed both `citation repair` behavior and `false refusal` cases. These results are consistent with the narrower claim that workflow-level diagnostics can make grounding, refusal, repair, and safety-boundary behavior more observable in the reported setup.

Data, Ethics, and Competing Interests

Data availability. The benchmark files, safety-stress suite, run metadata, and manuscript sources used in this diagnostic study are maintained as paper-facing artifacts. Any public release will exclude local absolute paths and unrelated project files.

Ethics statement. This work evaluates a software workflow on author-authored benchmark prompts and does not involve human-subject experiments, clinical deployment, medical diagnosis, or safety assurance for real training decisions.

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Responsible AI Disclosure

Generative AI tools were used for language-level editing and LaTeX formatting assistance during manuscript preparation. The authors remain responsible for the paper's claims, experimental design, reported results, citation use, and final submission checks.

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